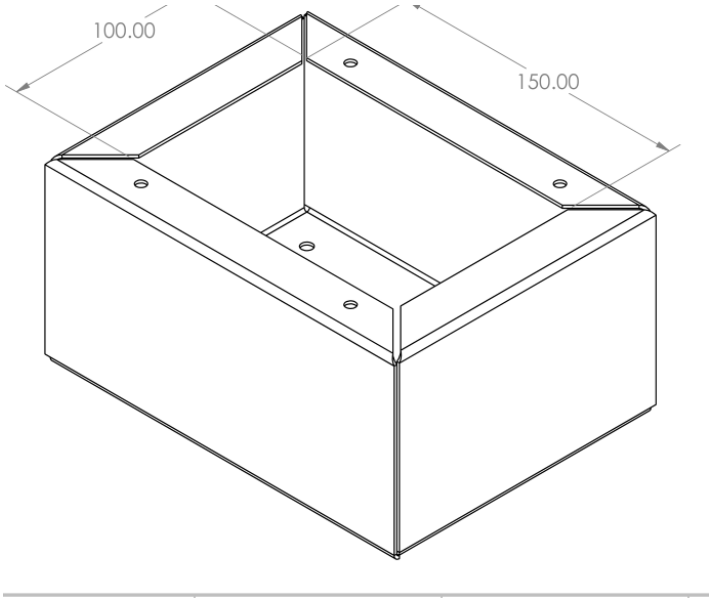
MECHANICAL DESIGN / SHEET METAL / CAD DOCUMENTATION

INDUSTRIAL SHEET METAL ENCLOSURE

A manufacturing-oriented CAD and documentation case study



10	1.5 mm	8	3
physical components	sheet thickness	body bends	A3 drawings

AUTHOR: REZA BABAPIRI

SOLIDWORKS 2026 / PRELIMINARY DESIGN REVIEW / NOVEMBER 2026

EDUCATIONAL PORTFOLIO — NOT A PRODUCTION RELEASE

01 / PROJECT INTENT

From CAD geometry to manufacturing documents

A self-directed mechanical CAD project exploring an enclosure made from a bent sheet-metal body, a removable lid, four self-clinching M4 nuts and four M4 socket-head screws. The emphasis is design traceability: a model, an assembly, dimensioned drawings and neutral manufacturing-oriented exports.

FORMED BODY	200 × 150 × 100 mm
SHEET MATERIAL	DC01 steel · EN 10130
THICKNESS	1.5 mm (both sheet parts)
BEND DETAIL	8 × 90° · inside R1.5 mm
LID SIZE	204 × 154 mm
BODY FLAT BOUNDS	439.19 × 389.19 mm
BODY MOUNTING	4 × Ø6 · 160 × 110 pitch
LID FASTENING	4 × Ø4.5 · 120 × 120 pitch

02 / CAD AND ASSEMBLY

Four-item bill of materials

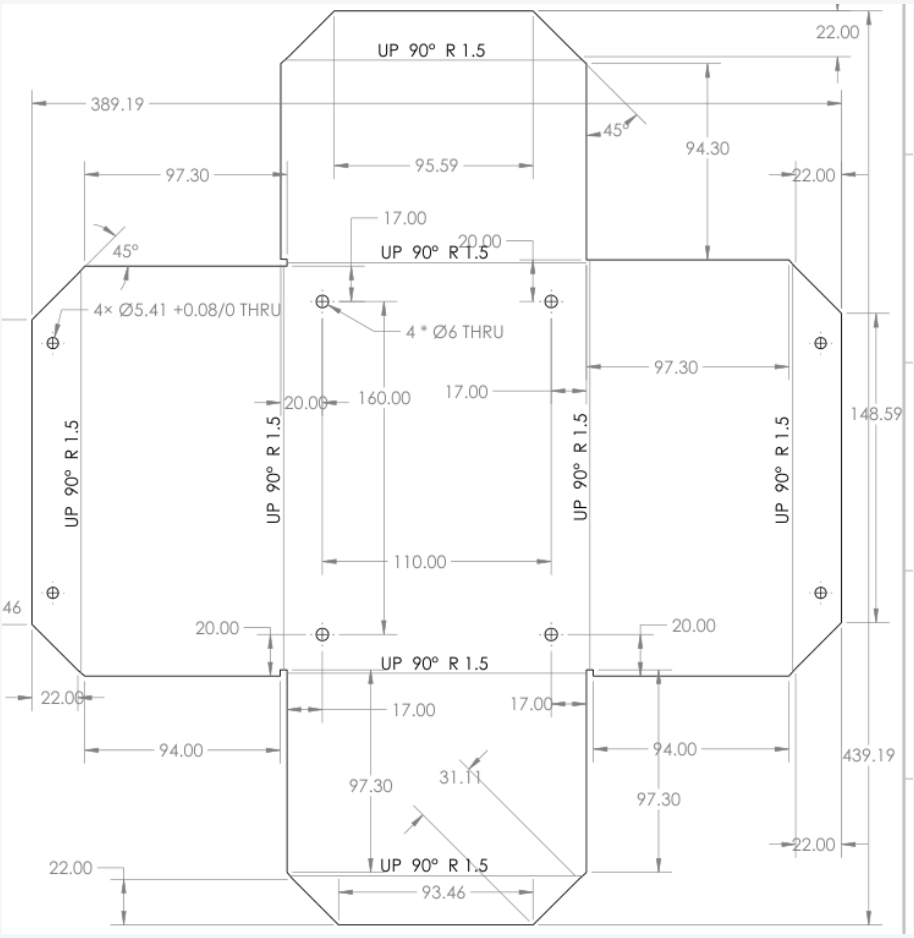
ITEM	PART	QTY
01	Formed enclosure body	1
02	Removable lid	1
03	S-M4-2ZC self-clinching nut	4
04	ISO 4762 M4 × 8 cap screw	4

DESIGN RATIONALE Inturned top flanges provide direct mounting faces for a removable lid, reducing the need for separate welded mounting tabs in this concept. Final clearance and clinch-nut installation must still be validated.

03 / SOURCE EVIDENCE

Sheet-metal development and hole geometry

The body drawing contains the formed isometric and a dimensioned developed blank. Eight 90° bend callouts and the four Ø6 mounting-hole pattern are documented on drawing IE-BDY-001. The four clinch-nut holes are Ø5.41 nominal in the final DXF.



EXCERPT / IE-BDY-001 — DEVELOPED BODY GEOMETRY

Nominal DXF inspection • Body: ~439.19 × 389.19 mm, four Ø6 circles and four Ø5.41 circles. • Lid: 204 × 154 mm, four Ø4.5 circles on a 120 × 120 mm pitch. Both exports use millimeter units and include educational-license text entities.

04 / DIGITAL VALIDATION

What was checked — and what remains open

Neutral CAD verification. The supplied STEP AP214 assembly was parsed using Open Cascade. It contains 10 solid bodies and all 10 passed the B-rep geometry validity check. This establishes digital topological integrity only; no structural or physical test was performed.

10 / 10
valid STEP solids

2
DXF cut profiles

3
A3 review drawings

RELEASE GATES BEFORE MANUFACTURE

CLINCHING INTERFACE

Confirm Ø5.41 (+0.08/0) design callout, thickness, edge distances and installation parameters against the exact PEM hardware datasheet.

PRESS-BRAKE PROCESS

Validate K-factor / bend deduction, bend sequencing, reliefs and tool access against the intended material and equipment.

DRAWING + BOM RELEASE

The new sheet corrects item 2 to Sheet Metal Lid, DC01. Define tolerances and finish; verify remaining screw part-number asterisk.

MASS CONSISTENCY

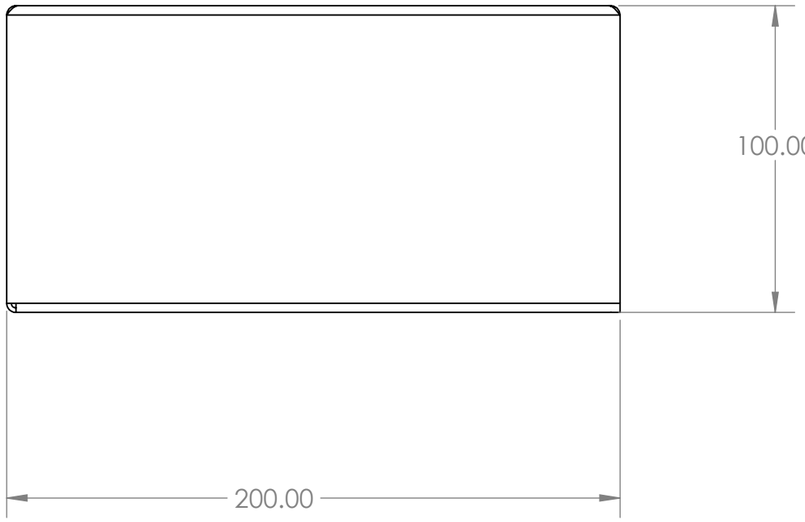
Reconcile reported assembly mass (1,663.03 g) with body (1,292.49 g), lid (369.17 g) and hardware masses.

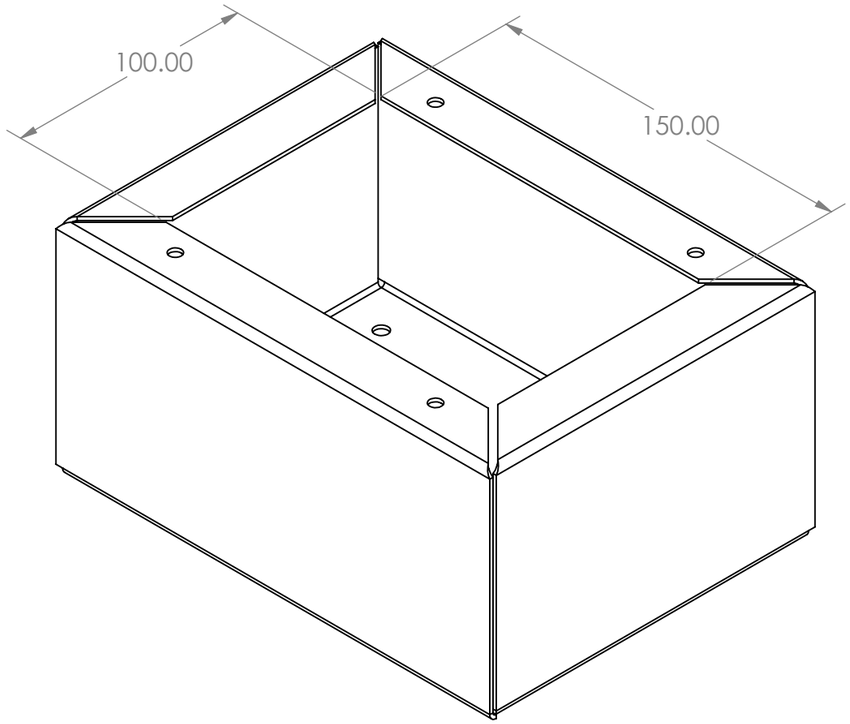
ASSEMBLY AND PROTOTYPE

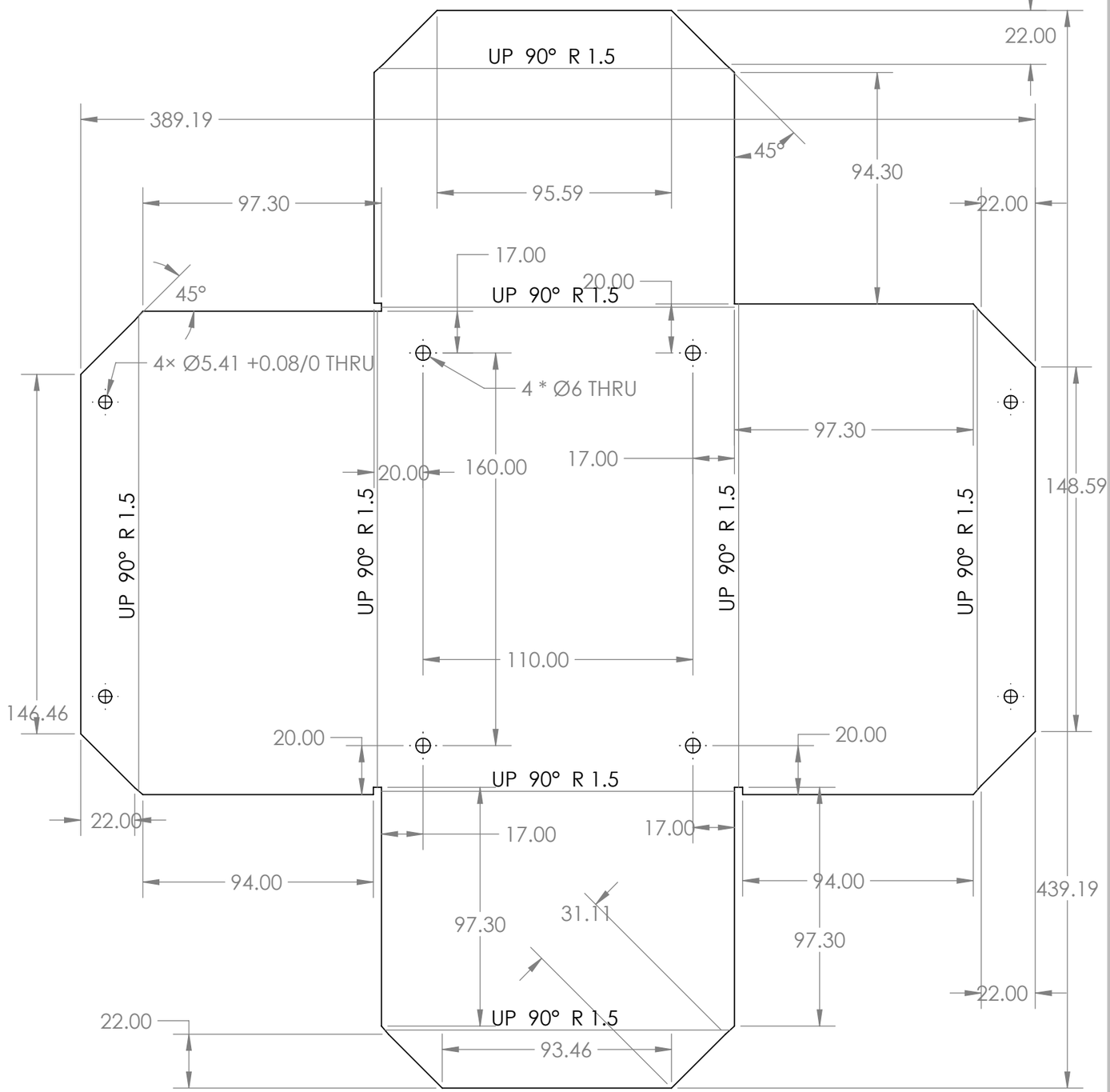
Verify fastener engagement, lid fit and any performance needs; no physical prototype or FEA results are claimed.

PUBLICATION AND LICENSING Drawings and DXF exports include “SOLIDWORKS Educational Product / For Instructional Use Only”. This case study is released as educational work and is not a commercial fabrication package or proof of a manufactured product.

SHEET THICKNESS: 1.5 mm



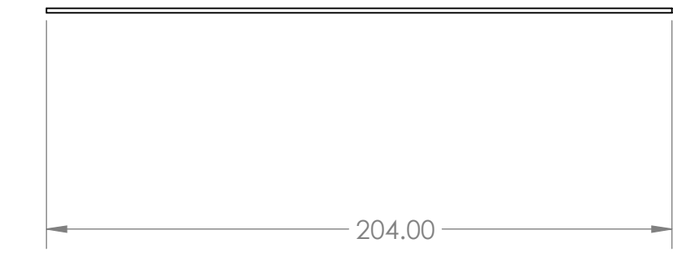


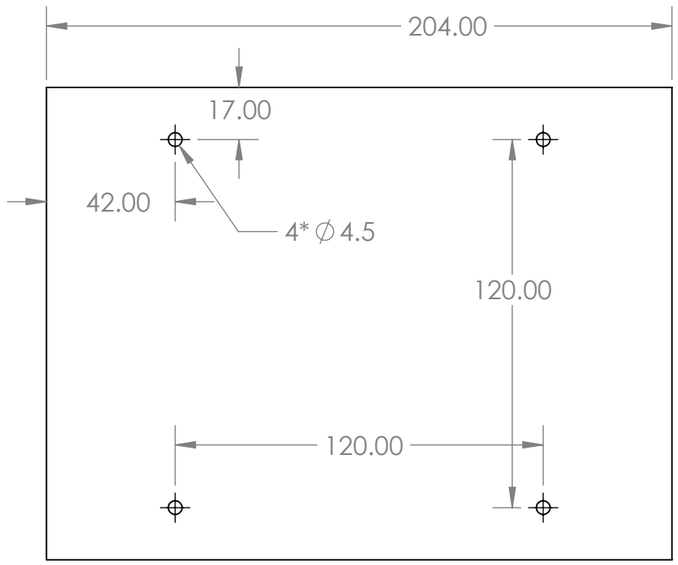


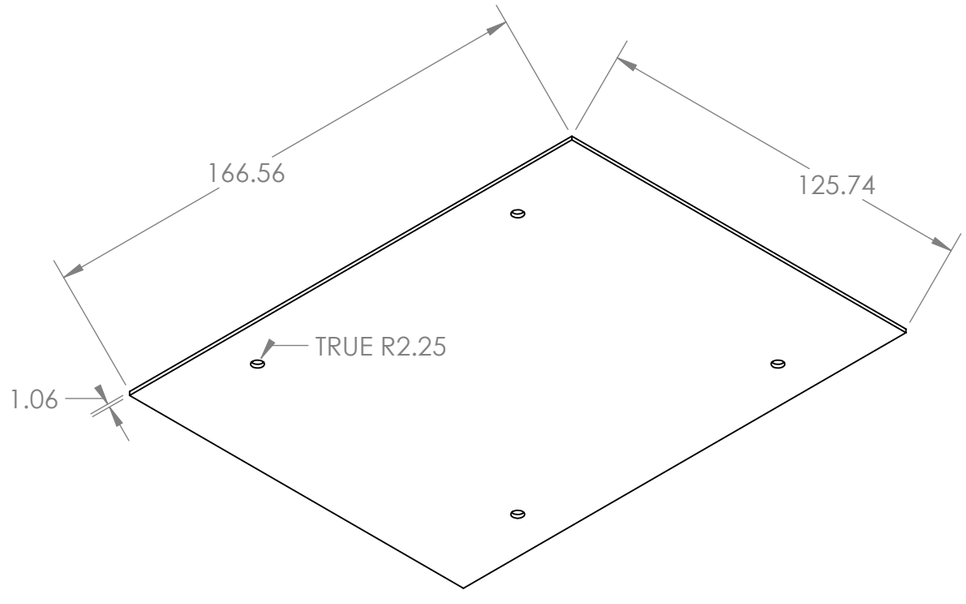
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TBD TOLERANCES: TBD LINEAR: TBD ANGULAR: TBD	FINISH: TBD	DEBURR AND BREAK SHARP EDGES	PRELIMINARY – FOR REVIEW	REVISION: R_00	
	NAME	SIGNATURE	DATE	TITLE:	
DRAWN	REZA BABAPIRI		2026-11-09	INDUSTRIAL ENCLOSURE – BODY	
CHK'D				DWG NO.	A3
APPROVED				IE-BDY-001	
MATERIAL: DC01 – EN 10130					
WEIGHT: 1292.49 grams				SCALE:1:2	SHEET 1 OF 1

SOLIDWORKS Educational Product. For Instructional Use Only.

SHEET THICKNESS: 1.5 mm

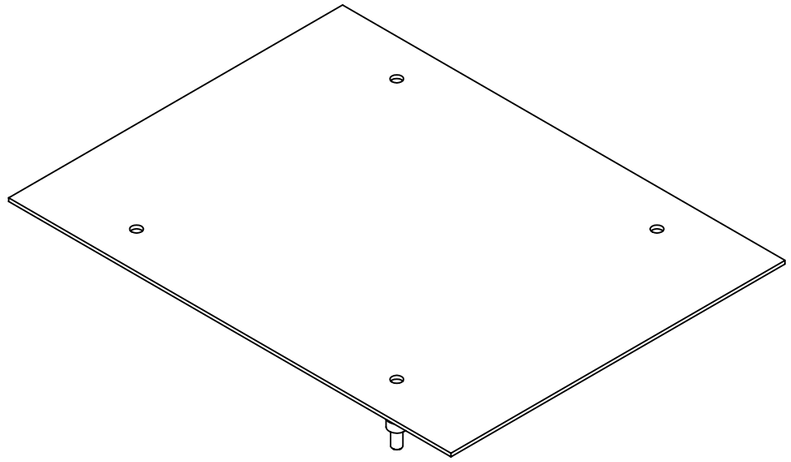


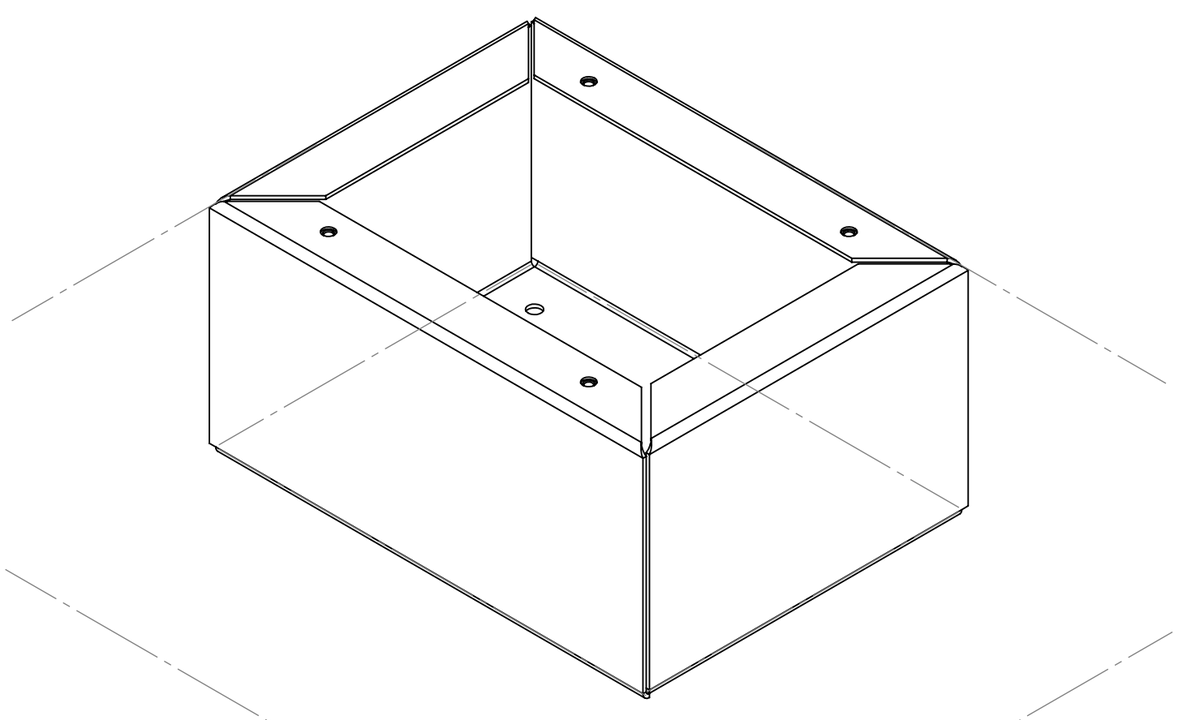


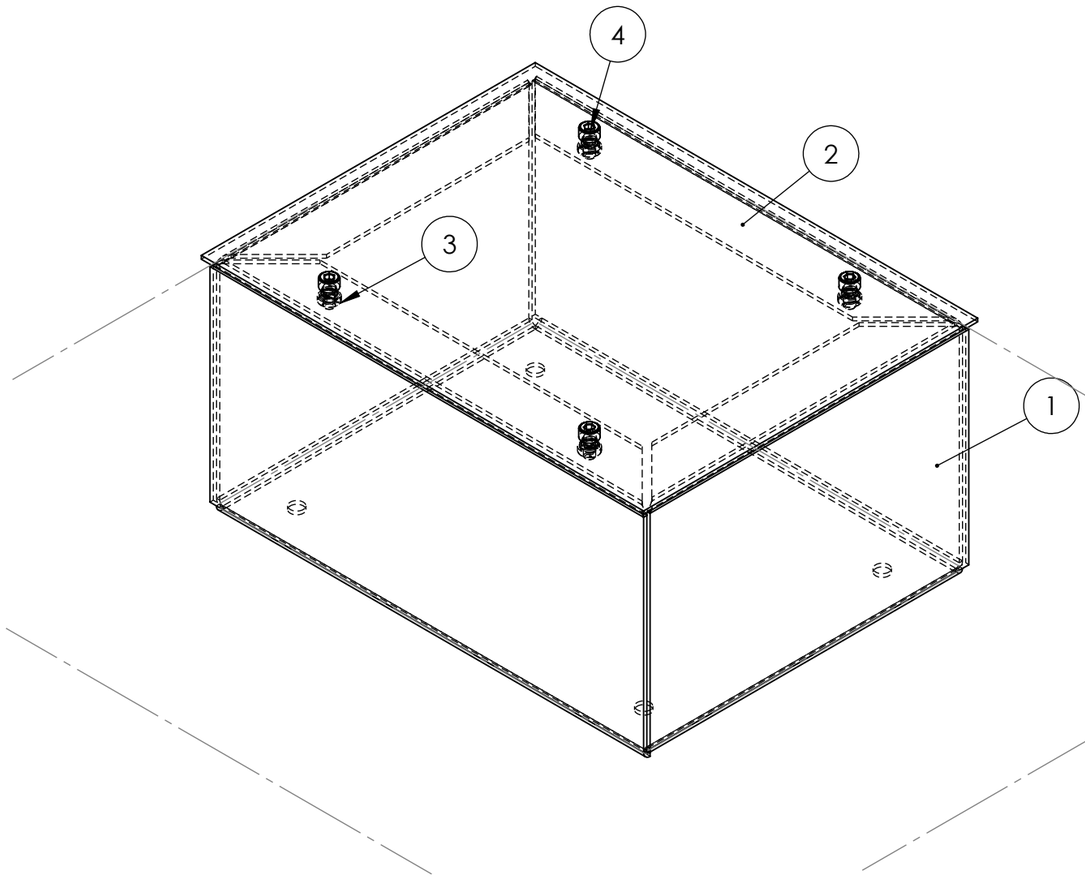


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TBD TOLERANCES: TBD LINEAR: TBD ANGULAR: TBD	FINISH: TBD	DEBURR AND BREAK SHARP EDGES	PRELIMINARY – FOR REVIEW	REVISION: R_00	
	NAME	SIGNATURE	DATE	TITLE:	
DRAWN	REZA BABAPIRI		2026-11-09	INDUSTRIAL ENCLOSURE – LID	
CHK'D				MATERIAL:	DWG NO.
APPROVED				DC01 – EN 10130	IE-LID-002 A3
				WEIGHT: 369.17 grams	SCALE:1:2 SHEET 1 OF 1

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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Industrial_Enclosure_Body	Sheet Metal Body, DC01	1
2	Industrial_Enclosure_Lid	Sheet Metal Lid, DC01	1
3	S-M4-2ZC	PEM Self-Clinching Nut, M4	4
4	ISO 4762 M4×8*	Socket Head Cap Screw	4

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TBD TOLERANCES: TBD LINEAR: TBD ANGULAR: TBD		FINISH: TBD		DEBURR AND BREAK SHARP EDGES	PRELIMINARY – FOR REVIEW	REVISION: R_00
	NAME	SIGNATURE	DATE			TITLE:
DRAWN	REZA BABAPIRI		2026-11-09			INDUSTRIAL ENCLOSURE – ASSEMBLY
CHK'D				MATERIAL:	DWG NO.	A3
APPROVED				SEE BOM	IE-ASM-003	
				WEIGHT: 1663.03 grams	SCALE:1:2	SHEET 1 OF 1

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